

CHAPTER 4

Designed and Adaptive Search for Failure

4

Search Is Choosing the Next Test

Once weakness can be measured, Chapter 4 asks how to spend the validation budget.

1 MEASURED WEAKNESS

Start from scores

Chapter 3 produced weakness measurements.

Now the question is: which candidate deserves the next evaluation?

2 FINITE BUDGET

Budget forces choices

We cannot test everything.

Search allocates evaluations toward informative failures, uncertainty and coverage gaps.

3 FIXED MODEL, CHANGING TESTS

The model stays fixed

Search changes the test configuration: region, medoids, prompt factors or comparator settings.

It does not validate by wishful retraining.

4 KEY PRINCIPLE

Search proposes where to look next. The evaluator provides the evidence.

A high discovery score is a reason to investigate.

Start With Deliberate Initial Coverage

Adaptive search needs a defensible starting population of candidates.

1 OBSERVED DATA

Twinning

For observed populations, select representative rows or anchors.

Good coverage before scores start steering the search.

2 CONTROLLABLE FACTORS

Design of Experiments

For generated or controllable tests, design factor combinations.

Space-filling designs avoid testing only familiar cases.

3 COVERAGE RECORD

What it establishes

Initial coverage shows what was examined.

It does not prove that all important failures have been found.

4 TEACHING POINT

Initial design is not the discovery result. It is the starting map for the search.

Search Methods Are Allocation Strategies

Different methods differ mainly in how they spend the next evaluation.

1 METHODS

Method	How the next test is chosen
Designed sweep	Run a preplanned batch before using feedback
Hill climbing	Try nearby candidates and move if one improves
Bandit	Allocate more trials to promising or uncertain arms
Evolution strategy	Mutate candidates, evaluate offspring and select
Bayesian optimization	Fit a surrogate and evaluate high-acquisition settings

2 COMMON THEME

All methods balance exploration, exploitation and cost. None replaces confirmation.

Evolution Strategy as the Search Engine

A reusable pattern for Discovery Loops: Parent → Mutate → Evaluate → Select → Repeat.

1 ES LOOP



2 WHY ES FITS VALIDATION

Works without gradients

The evaluator may be a notebook, a model run, a tool call or a human-reviewed scoring process.

Keeps tests executable

Mutation changes declared factors; construction checks reject invalid rows instead of inventing scores.

Supports portfolios

Selection can retain one best candidate or preserve Pareto tradeoffs across different weaknesses.

Important distinction: the optimizer proposes tests; the evaluator determines what the evidence supports.

Retain Useful Weaknesses, Recognize Limits

Search can refine within a representation; it cannot discover tests the representation excludes.

1 SEARCH VS RETENTION

Choosing the next evaluation is not the same as keeping the portfolio

Search decides where to spend budget next.

Frontier retention keeps distinct weaknesses, even when only one candidate receives the next trial.

2 REPRESENTATION LIMIT

More search cannot fix a missing representation

If the constructor cannot express a union of regions, a temporal factor or a multi-turn conversation, the optimizer cannot deliberately test it.

That is a structural problem.

3 BRIDGE TO CHAPTER 5

Chapter 4 searches the admitted space. Chapter 5 asks when the space itself must change.

This is the transition from optimization within a test space to proposing a revised test space.
